

The Magsail Cruiser

A Superconducting Magnetic-Sail Freighter for Propellantless Heliocentric Transport

Concept Study MS-CS-002 · Off-World Civilisation Toolkit · July 2026

Classification: Speculative engineering, constrained to known physics

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Abstract

The Sun exhales a permanent hurricane: the solar wind, a plasma stream of protons and electrons leaving the corona at 400-700 km/s and blowing outward through the entire solar system, day and night, for free. This study describes the Magsail Cruiser, a cargo vessel that sails that wind. The ship deploys a superconducting current loop kilometres in diameter, generating an artificial magnetosphere that deflects the wind's charged particles; every deflection transfers momentum, and the ship accelerates without carrying or expending any propellant whatsoever. We review the founding physics of the magnetic sail, the plasma-inflation technique that multiplies the sail's effective area far beyond its physical coil, the vessel's architecture, its operational envelope as a slow bulk freighter and as a braking system for incoming high-speed craft, and its incidental role as a crew radiation shelter. The concept requires no new physics; it waits principally on industrial quantities of high-temperature superconducting cable, a commodity whose production is already scaling for terrestrial fusion and power-grid markets.

1. Introduction: The Wind Was Always There

For the entire history of spaceflight we have crossed the solar system the expensive way, throwing reaction mass overboard and coasting on the ruinous arithmetic of the rocket equation. Meanwhile, an inexhaustible river of momentum has been flowing past every spacecraft ever launched. The solar wind carries roughly a nanopascal of dynamic pressure at Earth's distance from the Sun — feeble, but applied over a large enough area, and applied forever, feeble becomes formidable. Sailors on Earth's oceans understood this trade for five thousand years: engines are for harbours; the wind is for crossings.

A physical sail of fabric cannot catch a wind of protons — they would pass through any material sheet or destroy it. But charged particles obey magnetic fields, and a magnetic field weighs nothing. Andrews and Zubrin (1990) first showed that a large superconducting loop, carrying a persistent current and towed behind a spacecraft, deflects the solar wind around a magnetic bubble and receives its momentum in exchange; their follow-up analysis extended the idea to practical interplanetary transfer (Zubrin and Andrews, 1991). A decade later, Winglee et al. (2000) added the crucial economy measure: injecting a small quantity of plasma into the loop's field inflates the magnetosphere like breath into a balloon, so a coil hundreds of metres across can present an effective sail tens of kilometres wide. The Magsail Cruiser is the freighter these three papers have been waiting for.

2. Concept Description

The MS-1 'Aeolus' is an 18.5-tonne uncrewed tug built as a wheel without a rim's solidity: a central bus, eight kevlar rigging spokes under tension, and a hoop of high-temperature superconducting (HTS) cable two kilometres in diameter, deployed by slow spin and kept in shape by the magnetic hoop stress of

its own hundred-kiloampere persistent current. The bus houses the cryocoolers that hold the cable below its critical temperature, the plasma injectors that inflate the field, a modest solar array that powers both, and a cargo dock rated for twenty-tonne freight containers. There is no main engine. There is no propellant tank. The ship's entire propulsion system is a shape drawn in current.

Steering is done by geometry. Tilting the plane of the dipole relative to the wind vectors the thrust, in the same way a sailboat's keel and sail angle convert a following wind into progress across it; collapsing the field is the brake and the storm shelter. The vessel cannot sail directly sunward — the wind blows only outward — but by tacking against its own orbital motion it can lower its orbit, spiralling inward on a schedule, which is all a freighter requires (Zubrin and Andrews, 1991).

3. Physics of the Magnetic Sail

The thrust budget begins with the wind itself: at 1 AU the solar wind carries a proton density near five per cubic centimetre moving at roughly 450 km/s, a dynamic pressure of order 1-2 nPa (Zubrin and Andrews, 1991). A magnetosphere that presents an effective obstacle of radius R intercepts momentum across the area of that bubble, so an inflated magnetopause forty kilometres across gathers newtons of thrust — small, but continuous, and applied to a ship that spent none of its mass on engines. Over months, newtons compound into kilometres per second.

The decisive economics live in the inflation. A bare coil's dipole field falls off with the cube of distance, so the un-augmented magsail needs enormous coils to present a useful obstacle. Winglee et al. (2000) demonstrated in laboratory plasma chambers that seeding the field with injected plasma carries frozen-in flux outward and expands the magnetopause dramatically — the mini-magnetospheric plasma propulsion (M2P2) effect — buying effective area with grams per hour of argon rather than tonnes of copper. The same laboratory lineage, notably the diamagnetic-cavity experiments of Bamford et al. (2008), confirmed that such miniature magnetospheres do form and do exclude flowing plasma, which is also the basis of the sail's second profession: any crewed vehicle sheltering inside the bubble enjoys substantial deflection of solar energetic particles during storms.

Finally, the magsail owns one manoeuvre no rocket can copy economically: braking from interstellar speed. A vehicle arriving at a destination star at a few percent of light speed can deploy a magsail and decelerate against the destination's stellar wind and the interstellar medium itself, spending years but no propellant — the only known way to stop without carrying a second starship's worth of fuel (Perakis and Hein, 2016). Every beam-launched probe programme eventually rediscovers this and adds a magsail to its deceleration stage.

4. The Machine Itself

The pacing item is the hoop. Two kilometres of HTS cable must carry its persistent current at temperatures a cryocooler can sustain in sunlight — modern rare-earth barium copper oxide (REBCO) tape is superconducting below roughly 90 K and carries useful current density at 65-77 K, temperatures reachable with flight-proven cryocoolers and sunshielding rather than exotic helium plants. The cable is a layered architecture: REBCO tape stack at the core, a cryogenic circulation channel, structural jacket to carry the hoop stress, and multilayer insulation wrap. Quench — the sudden loss of superconductivity in a segment — is the failure mode that dominates the design; the loop is divided into independently valved thermal zones so a quench dumps its stored magnetic energy into one sacrificial radiator section rather

than propagating.

Deployment borrows from spin physics rather than robotics: the folded hoop is released from drums on the slowly rotating bus, centrifugal tension straightens the spokes, and the rising loop current then stiffens the ring magnetically — the sail rigs itself. Operations thereafter are almost embarrassingly quiet: the flight computer trims dipole tilt a few degrees per day, meters argon to the injectors, and watches space weather forecasts the way a schooner captain watched the barometer. Solar storms are not hazards but harvests; a coronal mass ejection multiplies the wind pressure tenfold, and the Aeolus surfs it.

5. Operations Concept

The Aeolus is not a passenger liner and never will be; its accelerations are measured in micro-gees. It is a bulk freighter for cargo that does not care when it arrives: water, propellant stock, structural metal, regolith shielding mass. Phase one is a technology demonstrator with a 100-metre coil in high Earth orbit, verifying inflation physics above the magnetopause where the true solar wind blows. Phase two is the first freight run: an outbound spiral from cislunar space to Mars orbit, eighteen to thirty months, delivering bulk consumables cheaper per kilogram than any chemical stage. Phase three integrates magsail braking stages into interstellar precursor probes and stations a rescue-shelter variant on the Earth-Mars corridor, offering any storm-caught crewed vessel a magnetic harbour.

Parameter	Phase 1 (Demo)	Phase 3 (Freighter)
Coil diameter	100 m	2,000 m
Loop current	10 kA	100 kA (persistent)
Inflated magnetopause	~2 km	~40-80 km
Thrust at 1 AU	millinewtons	1-5 N continuous
Propellant consumed	argon, g/hr (inflation only)	argon, g/hr (inflation only)
Cargo capacity	—	20,000 kg

Table 1. Representative parameters for the staged build-out.

6. Honest Difficulties

Four problems deserve candour. First, thrust direction: the wind blows outward only, so all inward transfers are indirect, slow, tacking spirals — the magsail is a freighter's tool, not a ferry's. Second, the wind is weather: solar wind pressure varies by factors of several on timescales of days, so arrival dates are forecasts, not promises. Third, superconductor mass and reliability at kilometre scale is unproven; a two-kilometre cryogenic machine with no possibility of repair visits must be engineered like a submarine cable, for decades of neglect. Fourth, the inflated-magnetosphere thrust levels measured by Winglee et al. (2000) remain debated at full scale — the honest status is that M2P2 multiplies sail area by a large but imprecisely known factor, and only a flight above the magnetopause will settle the number. Phase one exists to buy that answer.

7. Pathway to Reality

Nothing on the critical path is exotic. REBCO superconducting tape is in industrial production, its price falling annually under demand from compact fusion ventures and grid hardware. Flight cryocoolers

are standard on infrared astronomy missions. Plasma injection is laboratory-mature (Winglee et al., 2000; Bamford et al., 2008), and small-scale magnetoplasma sail demonstrations have been studied by multiple national agencies. A cubesat-class field-inflation experiment could fly within five years of funding; the 100-metre demonstrator within fifteen; the first freight spiral to Mars is plausibly a mid-century departure. The wind, meanwhile, continues to blow at no charge.

8. Conclusion

Chemical rockets sprint; the magsail walks — but it walks forever, carries no food, and never grows tired. Somewhere in the gap between those gaits lies the freight economy of the inner solar system: urgent things will always fly, and everything else will sail. The Aeolus is a bet that the oldest transport insight our species owns — that the crossing belongs to the wind — survives translation into vacuum. The rigging is current, the canvas is a magnetic field, and the trade winds leave the corona at four hundred kilometres per second, whether or not we choose to raise a sail into them.

References

Harvard referencing style (Cite Them Right).

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